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Saveetha Institute of Medical And Technical Sciences
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Smart Agriculture Farm Management System

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

CSA0211 – C Programming

to the award of the degree of

BACHELOR OF ENGINEERING

IN

BIOINFORMATICS

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DECLARATION

We, **Digvijay A, Reshini Priya B, Vishal J** of the Bioinformatics, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled '**Smart Agriculture Farm Management System with Crop Records, Resource Tracking, Harvest Management, and File-Based Data Processing**' is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place:

Date:

Signature of the Students with Names



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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Smart Agriculture Farm Management System with Crop Records, Resource Tracking, Harvest Management, and File-Based Data Processing**” has been carried out by **Digvijay A, Reshini Priya B, Vishal J** under the supervision of **Dr. K. Saravanan** and is submitted in partial fulfilment of the requirements for the current semester of the B.E **Bioinformatics** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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Signature With Student Name

ABSTRACT

Agriculture involves the continuous management of crop cultivation, farm resources, operational expenses, and harvest production. In many small-scale farming environments, these records may still be maintained manually, making it difficult to organize information, track expenses, monitor cultivation activities, and calculate production income accurately. Manual record keeping can also result in data duplication, calculation errors, and loss of important information. Therefore, a simple and efficient digital system is required to organize essential farm records in a structured manner.

The Smart Agriculture Farm Management System is a menu-driven application developed using standard C99 to provide an organized solution for basic agricultural record management. The system consists of three major modules: Crop Registration and Cultivation Management, Farm Resource and Expense Tracking, and Harvest Record and Production Management. The application uses core C programming concepts including arrays, strings, structures, functions, modular programming, input validation, and file handling.

The system allows users to add, view, search, update, and delete crop, resource, and harvest records. Data is stored permanently in text files such as crop.txt, resource.txt, and harvest.txt, allowing records to remain available even after the program is closed. A reporting component summarizes important farm information, including total expenses, harvest income, and net profit.

The project demonstrates how fundamental programming concepts can be applied to solve a practical agricultural management problem. By replacing scattered manual records with an organized digital system, the proposed solution improves data accessibility, reduces manual effort, and supports more efficient farm record management.

Table of Contents

S.No.	Topics	Page No
1	Introduction	7
2	Problem Identification and Analysis	9
3	Solution Design and Development	11
4	Results and Recommendations	15
5	Reflection on Learning and Personal Development	17
6	Conclusion	19
7	References	20
8	Appendices	21

List Of Figures and Tables

S.No.	Figure and Tables Title	Page No
1	Main Menu of the Smart Agriculture Farm Management System	21
2	Crop Registration & Cultivation Management Menu	22
3	Farm Resource & Expense Tracking Menu	22
4	Harvest & Production Management Menu	22
5	System Reports and Farm Performance Summary	23
6	Crop Data File (crop.txt)	23
7	Harvest Data File (harvest.txt)	23
8	User Data File (users.txt)	24
9	Three-Module System Architecture	24

CHAPTER 1: INTRODUCTION

1.1 Background Information

Agriculture is an essential sector that requires effective management of crops, resources, expenses, and production activities. A farm involves several stages, starting from crop registration and cultivation to resource utilization and final harvesting. Maintaining accurate records throughout these stages is important for understanding farm performance and making better management decisions.

In traditional farming practices, information such as crop details, planting dates, fertilizer usage, labour expenses, and harvest quantities may be maintained using notebooks or separate manual records. Managing many records in this manner can be time-consuming and may lead to calculation errors, duplication, difficulty in searching, or loss of information. A simple computerized system can help organize these records and make them easier to access and maintain.

The Smart Agriculture Farm Management System is developed as a menu-driven application using standard C99. The system provides a centralized method for maintaining essential farm information through three major modules: Crop Registration and Cultivation Management, Farm Resource and Expense Tracking, and Harvest Record and Production Management.

The project focuses on applying fundamental C programming concepts to a practical real-world problem. It uses arrays, strings, structures, functions, modular programming, input validation, and file handling to process and store farm data. The use of text files enables permanent storage of records without requiring an external database.

The system is designed to provide basic operations such as adding, viewing, searching, updating, and deleting records. It also performs calculations related to expenses and harvest income and generates summary information to support farm management.

1.2 Project Objectives

The primary objective of the project is to develop a simple and efficient software system for managing essential agricultural farm records.

The specific objectives are:

- To develop a digital system for maintaining farm records.
- To register and manage crop cultivation information.
- To track farm resources and operational expenses.
- To maintain accurate harvest and production records.
- To calculate total expenses and harvest income.
- To generate basic farm performance and profit reports.
- To provide permanent data storage using file handling.
- To implement CRUD operations such as add, view, search, update, and delete.

- To demonstrate the practical use of arrays, strings, structures, functions, and files in C programming.
- To develop the application using a modular and organized program structure.

1.3 Significance of the Project

The project is significant because it demonstrates how a simple software application can improve the organization of agricultural information. By maintaining crop, resource, expense, and harvest records digitally, the system reduces dependence on manual documentation and makes information easier to retrieve.

The project also provides an academic benefit by demonstrating how fundamental C programming concepts can be combined to develop a complete real-world application. The modular structure makes the system easier to understand, test, maintain, and extend.

1.4 Scope of the Project

The scope of the system is limited to basic farm record management. It includes:

- Crop registration and cultivation details
- Farm resource management
- Expense tracking
- Harvest and production records
- Income calculation
- Basic reporting
- File-based data storage
- Record search, update, and deletion

The current system does not include advanced features such as IoT sensors, weather forecasting, artificial intelligence, GPS monitoring, cloud storage, or database connectivity. These can be considered as future enhancements.

1.5 Methodology Overview

The project follows a modular development methodology. The overall requirements are first divided into three independent functional modules. Appropriate structures are then designed for storing crop, resource, and harvest information.

Each module is implemented using separate functions and source files. File-handling functions are used to save and retrieve records from text files. Input validation is incorporated to minimize invalid entries and duplicate records.

Finally, all modules are integrated through the main program, tested using different inputs and operations, and evaluated based on the system's ability to efficiently manage farm records and generate useful summaries.

CHAPTER 2: PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Farm management involves maintaining different types of information, including crop details, cultivation activities, resources, expenses, and harvest production. When these records are maintained manually, managing and retrieving information can become difficult, especially when the number of records increases.

Farmers may maintain crop information in one record while resource purchases, labour costs, and harvest details are recorded separately. This makes it difficult to obtain a complete view of farm activities. Manual calculations of total expenses, production income, and profit may also result in errors.

The main problem addressed by this project is the absence of a simple, centralized system for managing basic farm records efficiently. The proposed system addresses this problem by providing a structured digital platform for recording, processing, updating, and retrieving farm-related information.

2.2 Evidence of the Problem

Traditional paper-based record keeping requires continuous manual effort and can result in problems such as misplaced records, duplicate entries, incorrect calculations, and difficulty in searching historical information.

Digital agriculture has increasingly focused on using information and communication technologies to improve farm management and decision-making. Farm Management Information Systems can help organize information related to agricultural operations and support better management of farm activities.

However, advanced agricultural technologies may involve higher implementation costs and require technical infrastructure. Therefore, there is a need for simple and accessible solutions that can demonstrate the basic benefits of digital farm record management.

The proposed system addresses this requirement by providing a lightweight file-based application that does not depend on an external database or complex infrastructure.

2.3 Stakeholders

The major stakeholders associated with the proposed system are:

Farmers

Farmers are the primary users of the system. They can maintain crop, resource, expense, and harvest records in an organized manner.

Farm Managers

Farm managers can use the system to monitor resource usage, expenses, production, and overall farm performance.

Agricultural Students

The system can be used as an educational example to understand how programming can be applied to real-world agricultural problems.

Small-Scale Agricultural Organizations

Small farms or agricultural organizations can use the basic concept for maintaining structured records without requiring expensive software infrastructure.

Developers

The modular design provides developers with a foundation for adding advanced features such as database connectivity, web interfaces, IoT monitoring, and data analytics.

2.4 Supporting Data and Research

Digital technologies are increasingly being applied to agriculture for improving farm management, productivity, and decision-making. Farm Management Information Systems provide a structured approach to handling farm-related information and can integrate different areas of agricultural management.

Research also indicates that digitalization and automation can provide benefits to agricultural activities while challenges such as cost, infrastructure, and digital skills can affect adoption, particularly among smaller producers.

Based on these observations, the proposed project focuses on a simple, low-cost, and easy-to-understand digital record management system. Rather than attempting to replace advanced smart farming technologies, the project provides a basic foundation for managing essential agricultural information.

The problem analysis therefore establishes the need for a system that can:

- Centralize farm records.
- Reduce dependence on manual documentation.
- Improve accuracy of expense and income calculations.
- Provide faster access to stored information.
- Maintain records permanently.
- Provide a foundation for future smart agriculture features.

CHAPTER 3: SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The Smart Agriculture Farm Management System was developed using a modular programming approach. The project requirements were first analyzed and divided into three major functional modules: Crop Registration and Cultivation Management, Farm Resource and Expense Tracking, and Harvest Record and Production Management.

The data requirements for each module were identified and represented using C structures. Individual functions were then designed for operations such as adding, viewing, searching, updating, and deleting records.

The system was divided into multiple source and header files to maintain a clear separation of responsibilities. File-handling functions were implemented to provide permanent storage of records in text files. Input validation was also included to prevent invalid values, empty inputs, negative quantities, and duplicate identifiers.

After implementing the individual modules, they were integrated through the main program. The complete system was then tested using different inputs and operations to verify data storage, retrieval, modification, deletion, and report calculations.

3.2 Tools and Technologies Used

The project is developed using standard C99 without external C libraries.

Programming Concepts

- Arrays: Used for storing and processing collections of data.
- Strings: Used for names, dates, crop types, and other textual information.
- Structures: Used to represent crop, resource, and harvest records.
- Functions: Used to divide the program into reusable operations.
- File Handling: Used for permanent storage and retrieval of records.
- Modular Programming: Used to separate different system functionalities.
- Input Validation: Used to ensure reliable and valid user input.

Development Tools

- C99 Compiler / GCC – Compilation and execution.
- Text Files – Data storage.
- Console Interface – User interaction.

3.3 Solution Overview

The proposed system provides a centralized method for managing essential farm information through three main modules.

Module 1: Crop Registration and Cultivation Management

This module maintains information about crops cultivated on the farm. Users can add new crop records and perform view, search, update, and delete operations.

Important crop information includes:

- Crop ID
- Crop Name
- Crop Type
- Planting Date
- Expected Harvest Date
- Area Cultivated
- Season
- Farmer Name

The records are stored in crop.txt.

Module 2: Farm Resource and Expense Tracking

This module manages the resources required for farming activities and their associated expenses.

The system can maintain information related to:

- Seeds
- Fertilizers
- Pesticides
- Water
- Labour
- Machinery

The module provides resource management operations and calculates the total expenses incurred. The records are stored in resource.txt.

Module 3: Harvest Record and Production Management

This module manages information about harvested crops and production.

The system records:

- Harvest ID
- Crop ID
- Harvest Date
- Quantity Produced

- Unit
- Selling Price
- Total Income

The income is calculated based on the quantity produced and selling price. Harvest information is stored in harvest.txt.

3.4 System Architecture

The system follows a simple modular architecture:

User → Main Program → Functional Modules → File Handling → Text Files

The main program provides the menu-driven interface. Based on the user's selection, control is transferred to the appropriate module. Each module performs its operations and uses file-handling functions to store or retrieve data.

The major source files are:

- main.c
- crop.c
- resource.c
- harvest.c
- file.c
- utils.c

Header files are used to provide function declarations and improve communication between modules.

The reporting functionality reads information from the relevant files and generates summaries such as:

- Total number of crops
- Total expenses
- Total harvest
- Total income
- Net profit

3.5 Engineering Standards Applied

The project follows standard C99 programming practices and emphasizes structured and modular software development.

The following practices are applied:

- Modular source-code organization

- Separation of responsibilities
- Reusable functions
- Consistent naming conventions
- Input validation
- File error handling
- Structured data representation
- Maintainable code organization

These practices improve the readability, reliability, and maintainability of the application.

3.6 Solution Justification

A file-based C application was selected because the primary purpose of the project is to demonstrate the practical application of fundamental C programming concepts.

Using text files provides a simple method of permanent storage without requiring an external database. The modular structure also makes individual components easier to develop, test, debug, and maintain.

The proposed solution therefore provides an appropriate balance between technical simplicity and practical functionality, while creating a foundation that can later be extended into a database-driven, web-based, or IoT-enabled smart agriculture system.

CHAPTER 4: RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The Smart Agriculture Farm Management System successfully provides the core functionality required for managing basic farm records. The three major modules work together to organize crop, resource, expense, and harvest information.

The Crop Registration and Cultivation Management module enables users to add, view, search, update, and delete crop records. This provides a structured way to maintain cultivation information.

The Farm Resource and Expense Tracking module allows users to record resources such as seeds, fertilizers, pesticides, labour, and machinery expenses. The system can calculate and summarize the total expenses.

The Harvest Record and Production Management module records harvest information, including quantity produced and selling price, and calculates the corresponding income.

File handling provides permanent storage through crop.txt, resource.txt, and harvest.txt. Therefore, records can be retained and accessed even after the program is closed.

The reporting functionality combines relevant information to provide useful summaries such as total expenses, total income, and net profit. Overall, the system achieves the intended outcome of providing smart and efficient agricultural farm management through organized digital records.

4.2 Challenges Encountered

During the development of the system, several challenges were encountered.

File Handling

Implementing update and delete operations with text files required careful handling of existing records. Records sometimes need to be read and rewritten safely to maintain data consistency.

Input Validation

Handling incorrect inputs such as invalid IDs, empty strings, negative values, and incorrect menu choices required additional validation functions.

Modular Programming

Dividing the project into multiple source and header files required proper function declarations and communication between modules.

Data Consistency

Maintaining relationships between crop IDs and harvest records required consistent identification and storage of records.

These challenges were addressed through modular functions, validation mechanisms, structured data, and repeated testing.

4.3 Possible Improvements

Although the current system meets its primary objectives, several improvements can be introduced in future versions:

- Integration with a relational database.
- Web-based user interface.
- Mobile application support.
- User authentication and access control.
- Graphical dashboards and analytics.
- IoT-based soil and environmental monitoring.
- Weather information and alerts.
- Automated crop yield prediction.
- Cloud-based backup and synchronization.
- Automated harvest reminders.

These improvements could transform the current file-based application into a more comprehensive smart farming platform.

4.4 Recommendations

The system can initially be used as a lightweight farm record management solution. For larger farms with increasing amounts of data, the text-file storage approach can be replaced with a database system.

A future version could also integrate IoT sensors to automatically collect information such as soil moisture, temperature, and water usage. A web or mobile interface could make the system more accessible to farmers.

Advanced analytics and machine learning could eventually be incorporated to predict crop yield, analyze expenses, and provide data-driven recommendations. However, future development should continue to prioritize simplicity, affordability, reliability, and ease of use so that the system remains practical for small and medium-scale agricultural users.

CHAPTER 5: REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

Academic Knowledge

This capstone project helped me understand how fundamental C programming concepts can be applied to develop a complete real-world application. Concepts such as arrays, strings, structures, functions, and file handling were applied together rather than being studied individually.

The project also helped me understand how software can be designed to address practical problems in agricultural farm management.

Technical Skills

Through this project, I developed practical skills in:

- C99 programming
- Structures and arrays
- String handling
- Functions and modular programming
- File handling
- CRUD operations
- Input validation
- Data processing
- Debugging and testing

Working with multiple source and header files also improved my understanding of organizing a larger C program.

Problem-Solving and Critical Thinking

The project required breaking a broad farm management problem into smaller and manageable components. Dividing the system into crop, resource, and harvest modules made the development process more organized.

Implementing file-based CRUD operations also required logical thinking because records had to be maintained correctly while performing search, update, and delete operations.

5.2 Challenges Encountered and Overcome

One of the main challenges was implementing reliable file handling. Reading and writing records was relatively straightforward, but updating and deleting existing records required careful processing of the stored data.

Another challenge was input validation. Users may enter invalid values, duplicate IDs, empty strings, or incorrect menu choices. These situations were handled using dedicated validation functions.

Organizing the application into multiple source files was also a learning experience. Proper use of header files and function declarations was required to ensure that the different modules worked together correctly.

Overcoming these challenges improved my debugging skills, logical thinking, and confidence in developing larger C programs.

5.3 Application of Engineering Standards

The project follows structured programming and standard C99 practices. The application is divided into independent modules, with functions designed for specific responsibilities.

Proper naming, modular source files, input validation, file error handling, and reusable functions were used to improve code quality. These practices helped make the system easier to understand, test, maintain, and extend.

Although the project is not formally certified against a specific engineering standard, standard programming practices were followed throughout the implementation.

5.4 Insights into the Industry

This project provided an understanding that developing a useful software system involves more than writing code. Requirement analysis, system design, data organization, testing, error handling, and documentation are equally important.

The project also provided insight into how software can be applied to agriculture. A basic record management system can serve as a foundation for more advanced solutions involving databases, web applications, IoT devices, cloud platforms, and data analytics.

The experience demonstrated the importance of designing technology around practical user requirements rather than focusing only on technical features.

5.5 Conclusion of Personal Development

Overall, the capstone project contributed significantly to my technical and personal development. It strengthened my understanding of C programming and provided practical experience in designing and implementing a complete software application.

The project improved my ability to analyze problems, divide them into manageable modules, debug errors, and develop solutions systematically. It also increased my confidence in applying programming knowledge to real-world problems.

The experience provides a strong foundation for exploring advanced areas such as database management, web development, IoT, data analytics, and intelligent agricultural systems in future projects.

CHAPTER 6: CONCLUSION

The Smart Agriculture Farm Management System was developed to provide a simple and organized solution for managing essential agricultural records. The project addresses the difficulties associated with maintaining crop, resource, expense, and harvest information manually.

The system consists of three major modules: Crop Registration and Cultivation Management, Farm Resource and Expense Tracking, and Harvest Record and Production Management. These modules provide operations for adding, viewing, searching, updating, and deleting records. File handling is used to store the information permanently in crop.txt, resource.txt, and harvest.txt.

The project successfully demonstrates the practical application of fundamental C99 programming concepts, including arrays, strings, structures, functions, modular programming, input validation, and file processing. The reporting functionality further improves the system by providing summaries of total expenses, income, and net profit.

Although the current implementation is a lightweight file-based application, its modular structure provides a strong foundation for future improvements. Database connectivity, web interfaces, IoT-based monitoring, cloud storage, data analytics, and machine learning can be integrated in future versions.

Overall, the project achieves its intended outcome of smart and efficient agricultural farm management while providing practical experience in software design, C programming, file management, problem-solving, and system development.

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Appendices

Appendix A- Project Source Code Structure

Smart Agriculture Farm Management System

- main.c
- crop.c
- crop.h
- resource.c
- resource.h
- harvest.c
- harvest.h
- file.c
- file.h
- utils.c
- utils.h
- crop.txt
- resource.txt
- harvest.txt

Appendix B- System Modules

```
=====
Smart Agriculture Farm Management System
=====
1. Crop Registration & Cultivation Management
2. Farm Resource & Expense Tracking
3. Harvest Record & Production Management
4. Reports
5. Exit
-----
Enter your choice (1-5): |
```

Fig 1: Main Menu of the Smart Agriculture Farm Management System

```
=====
Crop Registration
=====
1. Add Crop
2. View All Crops
3. Search Crop
4. Update Crop
5. Delete Crop
6. Back to Main Menu
-----
Enter choice: |
```

Fig 2: Crop Registration & Cultivation Management Menu

```
=====
Farm Resource & Expense Tracking
=====
1. Add Resource
2. View All Resources
3. Update Resource
4. Delete Resource
5. Back to Main Menu
-----
Enter choice: |
```

Fig 3: Farm Resource & Expense Tracking Menu

```
=====
Harvest & Production Management
=====
1. Add Harvest Record
2. View All Harvests
3. Delete Harvest Record
4. Back to Main Menu
-----
Enter choice: |
```

Fig 4: Harvest & Production Management Menu

System Reports	
Metric	Value
Total Crops Registered	6
Total Resources Tracked	6
Total Harvest Records	3
Total Expense	\$10820.00
Total Income	\$181500.00
Net Profit	\$170680.00
Status: PROFITABLE (Good job!)	
Press Enter to continue...	

Fig 5: System Reports and Farm Performance Summary

Appendix C – Data Files

crop.txt							
101	Wheat	Cereal	2025-10-15	2026-04-10	50.5	Winter	John Smith
102	Corn	Grain	2026-04-01	2026-09-15	75.0	Summer	David Miller
103	Soybeans	Legume	2026-05-10	2026-10-20	40.0	Summer	Sarah Jones
104	Tomatoes	Vegetable	2026-03-15	2026-08-01	15.5	Spring	Carlos Ruiz
105	Cotton	Fiber	2026-04-20	2026-11-05	60.0	Summer	Emma White
1	Wheat	Summer	12-11-2006	24-04-2007	3.00	summer	vijay

Fig 6: Crop Data File (crop.txt)

harvest.txt							
301	101	2026-04-15	150000.00	kg	0.35	52500.00	
302	104	2026-08-05	25000.00	kg	1.80	45000.00	
303	102	2025-09-20	300000.00	kg	0.28	84000.00	

Fig 7: Harvest Data File (harvest.txt)

```

users.txt
File Edit View H1 B I S G ...
901|admin|admin123|admin@smartfarm.local|Admin|Active
902|johndoe|farmerpass|john.d@smartfarm.local|Manager|Active
903|swilliams|work123|sarah.w@smartfarm.local|Worker|Active
904|inspector|viewonly|audit@agrigo.local|Viewer|Active
905|cruiz|tomato45|carlos@smartfarm.local|Worker|Inactive
9799|digvijay|password|digvijay8c@gmail.com|Worker|Active

```

Fig 8: User Data File (users.txt)

Appendix D – System Flow

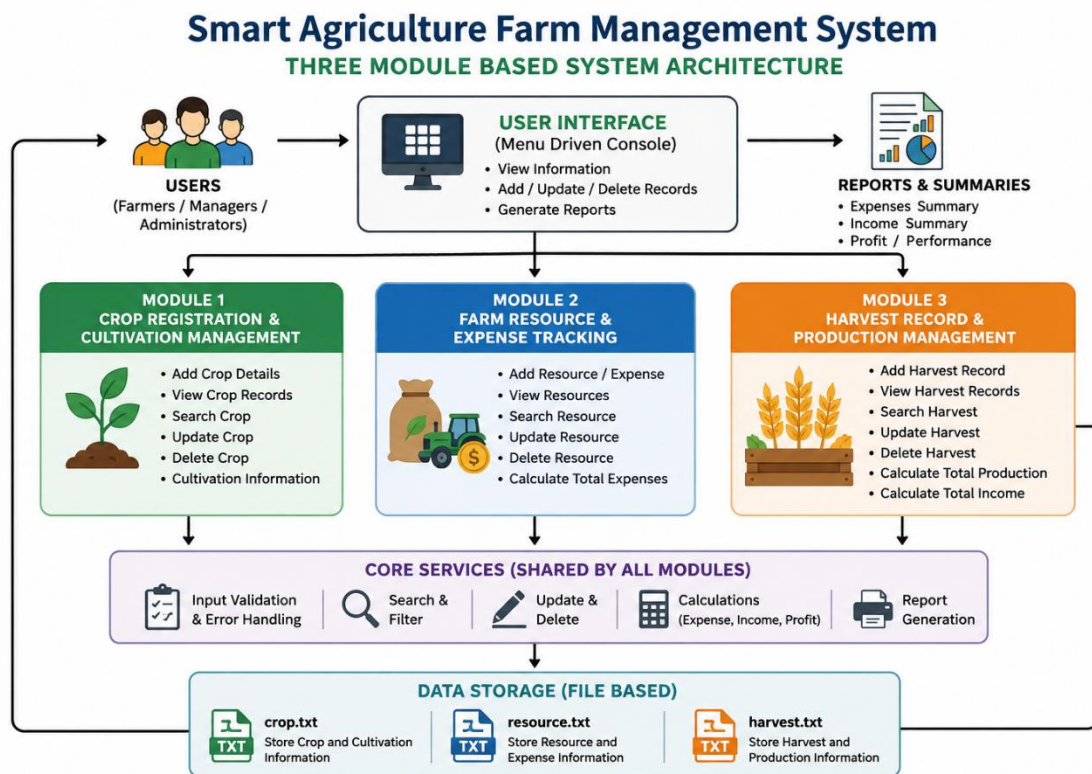


Fig 9: Three-Module System Architecture